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Do concurrent task demands impact the benefit of automation transparency?☆

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ABSTRACT

Automated decision aids typically improve decision-making, but incorrect advice risks automation misuse or disuse. We examined the novel question of whether increased automation transparency improves the accuracy of automation use under conditions with/without concurrent (non-automated assisted) task demands. Participants completed an uninhabited vehicle (UV) management task whereby they assigned the best UV to complete missions. Automation advised the best UV but was not always correct. Concurrent non-automated task demands decreased the accuracy of automation use, and increased decision time and perceived workload. With no concurrent task demands, increased transparency which provided more information on how the automation made decisions, improved the accuracy of automation use. With concurrent task demands, increased transparency led to higher trust ratings, faster decisions, and a bias towards agreeing with automation. These outcomes indicate increased reliance on highly transparent automation under conditions with concurrent task demands and have potential implications for human-automation teaming design.

1. Introduction

Automated decision aids are increasingly used in the workplace (Vagia et al., 2016). For example, in defence and security settings, decision aids make recommendations to operators regarding the deployment of uninhabited vehicles (UVs; Calhoun et al., 2018), and in air traffic control, separation assurance decision aids recommend aircraft conflict interventions (Trapsilawati et al., 2016). Although automation in such contexts tends to be highly reliable, it can potentially provide incorrect advice. If operators do not appropriately verify automated advice or do not otherwise understand the rationale underlying advice, they may accept incorrect advice (automation *misuse*) or reject correct advice (automation *disuse;* Lee and See, 2004; Parasuraman and Riley, 1997), with potentially serious consequences.

Misuse and disuse of automation depends on factors such as operator trust in automation (Lee and See, 2004). The formation of trust in automation has been likened to human-human trust formation (Madhavan and Wiegmann, 2007b). However, subtle differences (e.g., the relative source credibility/reliability of advisors) influence the way in which humans trust automated aids compared to human aids (Lyons and Stokes, 2012; Madhavan and Wiegmann, 2007a; Pearson et al., 2019). Much like trust between humans, trust in automation can develop based on the information provided to the operator regarding *how* the automation functions (Dzindolet et al., 2003; Lee and See, 2004). Providing this form of information is generally referred to as *automation transparency*.

1.1. Automation transparency

Increasing automation transparency (aka visibility; Dorneich et al., 2017, or observability; Christoffersen and Woods, 2002) is designed to assist operators in understanding the rationale underlying automated advice and what to expect if advice is followed. However, because increased transparency invariably involves presenting *more* information to operators, a key consideration is how best to design transparency to increase the accuracy of automation use without increasing decision time or workload.

Two main theoretical models have been used to guide transparency design; the Human-Robot Transparency Model (Lyons, 2013) and the Situation-Awareness Agent-Based Transparency model (SAT; Chen

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et al., 2014). In Lyons' model, four factors are described, including intentional (why and for what purpose the automation was created), task (automation goals and progress, awareness of errors), analytical (rationale for behaviour), and environmental (environment dynamics) factors. The SAT model outlines three levels of transparency including the automation's goals, intentions, and proposed actions (Level 1), reasoning behind advice (Level 2), and projected future outcomes if advice is followed and any uncertainty associated with those outcomes (Level 3).

Research has examined transparency in a variety of task domains (see Bhaskara et al., 2020, and Van de Merwe et al., 2022, for reviews), and increased transparency often allows humans to use automated advice more accurately (i.e., correctly accept/reject advice). However, some studies reported that transparency did not improve automation use (Guznov et al., 2020; Roth et al., 2020), or degraded it due to a strong bias to agree with automation (Bhaskara et al., 2021; Bussone et al., 2015). Further, while some studies found that increased transparency reduces or does not affect decision time (e.g., Skraaning and Jamieson, 2021), others reported slower decision times (e.g., Stowers et al., 2016). No consistent evidence indicates that transparency increases perceived workload, with some studies reporting a reduction in workload (Panganiban et al., 2020; Skraaning and Jamieson, 2021). Operator trust and perceived automation usability can increase with increased transparency, although findings are inconsistent (Bhaskara et al., 2020).

The task domain used in the current study was a UV management task. A number of prior studies have examined automation transparency in UV tasks due to its relevance to Defence contexts (e.g., Bhaskara et al., 2021; Loft et al., 2022; Mercado et al., 2016; Stowers et al., 2020). Participants in these studies were required to send the optimal UV to complete missions by either accepting the automation's recommended UV (Plan A) or by rejecting the advice and choosing the alternative UV (Plan B). Participants were either provided with low (e.g., basic information, plans/intent), medium (e.g., automation reasoning), or high (e.g., projected outcome, uncertainty) transparency. Increased transparency generally led to more accurate automation use (or a decreased need to verify advice; Loft et al., 2022), with no cost to decision time or workload (Bhaskara et al., 2020; Mercado et al., 2016; Stowers et al., 2020; but see Bhaskara et al., 2021). Furthermore, trust and usability ratings have sometimes been higher with increased transparency (Mercado et al., 2016; Stowers et al., 2020).

Overall, UV research indicates that increased transparency, more often than not, improves outcomes, but has the potential to increase automation bias and degrade the accuracy of automaton use, or increase decision time. However, studies to date have only required participants to complete one primary task. This is a limitation because UV operators are often required to complete concurrent tasks that do not have automation assistance. It is therefore important to examine the extent to which the benefits of transparency remain in the presence of concurrent task demands competing for attention, because as reviewed below, the manner in which humans monitor automation, and thus process transparency, could vary.

1.2. Concurrent task demands and automation transparency

A common assumption in the cognitive science literature is that humans have a fixed amount of attentional capacity (cognitive resources), and that concurrent tasks compete for these limited-capacity cognitive resources (Kahneman, 1973; Navon and Gopher, 1979; Norman and Bobrow, 1976). It follows then that concurrent non-automated task demands have been consistently shown to increase automation misuse (e.g., Bailey and Scerbo, 2007; Karpinsky et al., 2018; Rovira et al., 2002). For example, Parasuraman et al. (1993) found that participants completing concurrent tasks were poorer at detecting automation failures compared to those performing a single automation-assisted task. Bagheri and Jamieson (2004; Metzger and Parasuraman, 2005) replicated this, with eye-tracking data indicating that participants sampled automation-assisted task displays less frequently than non-automated task displays. Allocating attention to non-automated tasks at the expense of automation-assisted tasks could be viewed as an adaptive strategy to free up cognitive capacity, at the expense of tasks that automation can routinely handle reliably (Kaber, 2018; Moray and Inagaki, 2000), but this strategy risks increasing automation misuse.

We examined the extent to which any observed benefits of increased automation transparency in a UV task context remain with concurrent task demands. The benefits of transparency on automation use may be null/minimal if individuals do not allocate the attention required to adequately process transparency information to verify automated advice. Supervisory control theories assume that human information-seeking behaviour is sensitive to the same factors that determine mathematical optimality (Moray, 1986; Senders, 1983), and thus the probability that an individual will attend to transparency information whilst performing concurrent tasks should depend on the relative expected value (relevance) and cost of accessing that information (Fu and Gray, 2006; Wickens et al., 2015). A strong bias toward agreeing with automated advice (e.g., Bhaskara et al., 2021; Bussone et al., 2015) may decrease expected value, and transparency needs to be appropriately designed to minimize perceived information access costs.

To our knowledge, only one study has examined the impact of transparency with concurrent task demands (Wright et al., 2017). They found that increased transparency did not benefit accuracy of automation use, or impact any other outcomes (e.g., decision time), when there were concurrent non-automated task demands in a military convoy task. While informative, this study was substantially underpowered statistically.

1.3. Current study

We examined the impact of increased transparency on the accuracy of automation use, decision time, subjective workload, and perceived trust/usability, both with and without concurrent non-automated task demands. Under concurrent task conditions, participants were instructed that the "mission" and "image analysis" tasks were equally important. The mission task required participants to select the optimal UV based on vehicle capabilities (e.g., time to destination), mission-specific weightings of those capabilities, and environmental factors that impact capabilities. Automated assistance (the *Recommender*) evaluated each UV to advise the optimal UV to complete each mission. Participants were instructed that the Recommender was highly reliable, but not perfect. Participants decided whether to agree with automated advice and select Plan A, or instead choose Plan B.

Participants received either low, medium, or high transparency. All provided participants with information regarding how the Recommender evaluated the importance of each UV capability based on weightings. Medium transparency further indicated which UV the Recommender considered to be "better" and "poorer" on each capability, as well as a visualization of calculated capability scores for each UV. High transparency further indicated how the Recommender calculated its UV capability scores - outlining environmental factors taken into consideration and their projected impact on UV capabilities. High transparency was designed as the benchmark condition, and as such we examined the impact of high compared to low transparency, and high compared to medium transparency.

The concurrent task was an image analysis task, requiring participants to detect icon changes on an aerial map, emulating situations in which operators analyse images from UVs to detect threats (e.g., Abich IV et al., 2017). A mixed design was used, with the between-subjects factor of transparency (low, medium, high) and the within-subjects factor of concurrent task (present, absent). Automation was 80% reliable. This reliability was chosen because it was above the 70% reliability threshold suggested by Wickens and Dixon (2007) required to ensure humans at least partially rely on automated advice, and because it provided enough data points to examine differences in correct rejection rates as a function of condition.

Transparency should increase the accuracy of automation use when there is no concurrent task, with more accurate use with high compared to medium, and high compared to low transparency. Less accurate automation use, slower decisions, and increased perceived workload were expected with concurrent task demands, and the key question concerned the extent to which increased transparency would improve automation use with concurrent task demands. Increased transparency may increase perceived trust and automation usability.

2. Methods

2.1. Participants

Participants were 212 (141 females, 70 males, 1 non-binary; M = 21.66 years) undergraduates who took part for course credit and received a performance incentive of up to AUD$30. Participants were randomly assigned to either low (N = 71), medium (N = 71), or high (N = 70) transparency conditions. This research complied with the American Psychological Association Code of Ethics and was approved by the Human Research Ethics Office at The University of Western Australia. Informed participant consent was obtained.

2.2. Uninhabited vehicle simulation

The UV simulation was designed in consultation with Australian Defence and was presented on two desktop monitors (left monitor = mission task; Fig. 1, right monitor = image analysis task; Fig. 3). The mission task had 100 trials, with 50 trials completed concurrently with the image analysis task.

2.2.1. Mission task

Participants were required to deploy the optimal UV to complete each mission. Mission descriptions were presented in the mission window. There were 50 descriptions, each being used twice (Appendix A and B). Mission descriptions were irrelevant to which UV was optimal (only used for face validity). The tactical map provided an aerial view of rural, coastal, or urban areas. The tactical map displayed the location of the search area (translucent black box), and two UVs which could either be ground (UGV), aerial (UAV), or surface (USV) vehicles. For each mission, UVs were assigned a unique colour (blue or purple) and were numbered 1 or 2, in no particular order. Attached to each UV was a line depicting its path to reach the search area.

The optimal UV was determined based on three attributes: vehicle capabilities (time to destination, discoverability, fuel consumption); capability weightings; and environmental factors that impact capabilities. Time to destination (value presented next to timer symbol in Fig. 1) referred to how long the UV would take to reach the search area (lower = quicker). Discoverability (binocular symbol) referred to how discoverable the UV was by third parties (lower = less discoverable). Fuel consumption (fuel gauge symbol) referred to how much fuel the UV would consume to reach the search area (lower = less fuel). These capabilities were presented next to each UV on the tactical map (Fig. 1).

Each UV capability was assigned a weighting (percentage), presented in the weightings window (Fig. 1). Higher weightings indicated more important capabilities. Each mission used one of five weighting sets that varied in difficulty (Table 1). On some trials, where two capabilities had equal weightings, participants needed to consider a combination of capabilities. For example, if time to destination was weighted 45%, discoverability 45%, and fuel consumption 10%, it was necessary to consider which UV scored lower on two capabilities (e.g., lower on both discoverability and fuel consumption). On other trials, it was only necessary to consider one large weighting. For example, if time to

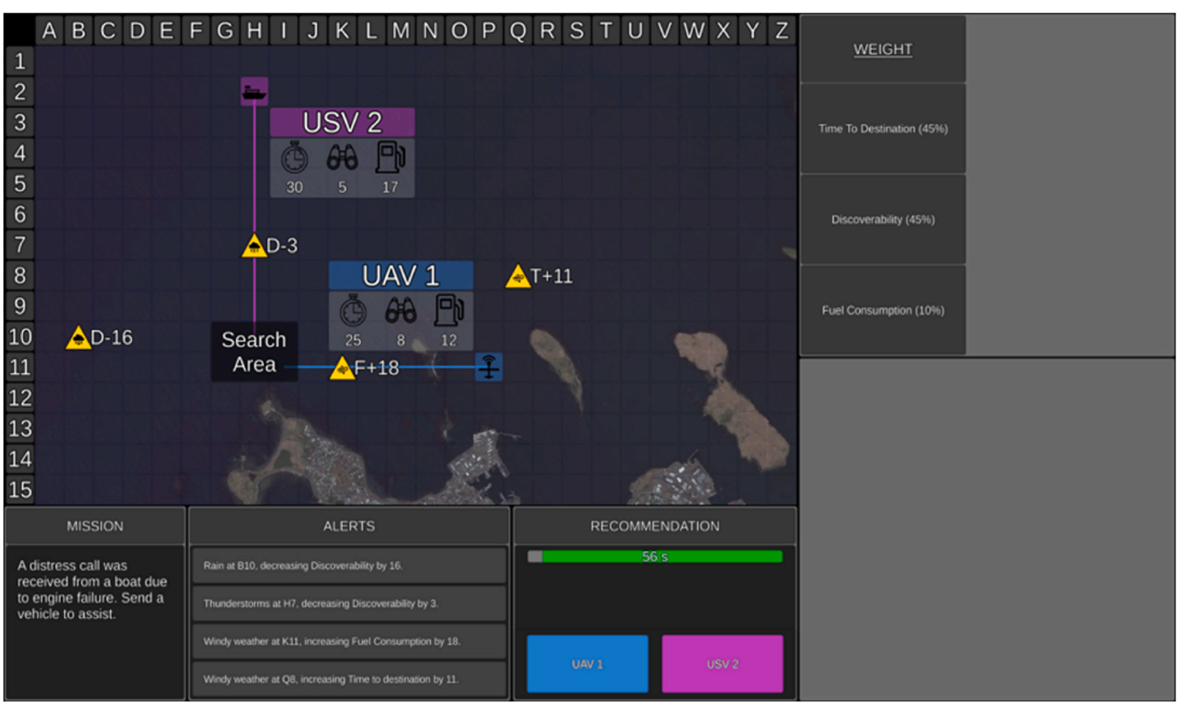


Fig. 1. *Mission task with no automated (Recommender) assistance (manual blocks).*
Note. The mission task display included the tactical map, weightings, mission, alerts, and recommendation windows. Presented on the tactical map was the location of the search area (translucent black box), two UVs, capabilities for each UV (time to destination, discoverability, and fuel consumption), the path each UV would take to reach the search area, and environmental hazard symbols. Relevant environmental factors were placed on the paths of UVs. The map in this example is a coastal area. The weightings window presented the weighting assigned to each capability. Mission objectives for each trial were presented in the mission window. Environmental factors and their impact were also presented as messages in the alerts window. The recommendation window indicated the time remaining and the two plan selection buttons, one for each UV.

Table 1
Capability weighting combinations and their associated level of difficulty.

Weighting Combination	Difficulty
40, 40, 20	Hard – need to consider multiple capabilities
45, 45, 10	Hard – need to consider multiple capabilities
40, 30, 30	Hard – need to consider multiple capabilities
80, 10, 10	Easy – can consider one capability
60, 20, 20	Easy – can consider one capability

destination was weighted 10%, discoverability 80%, and fuel consumption 10%, participants only needed to consider which UV scored lower on discoverability.

On each trial, four environmental factors were presented on the tactical map as yellow hazard symbols, which could be irrelevant or relevant to UV capabilities. Relevant factors could positively or negatively impact vehicle capabilities. The hazard symbols represented the type of factor (e.g., rain), location, capability impacted (T = time to destination, D = discoverability, F = fuel consumption), direction of impact (+= positive, - = negative), and level of impact. When factors were relevant, the hazard symbol appeared on the path of a UV (Fig. 1). Irrelevant factors were not presented on a UV path and could be ignored. Four messages were also presented in the alerts window regarding the factors and their impact. On each trial there could be between zero and three relevant factors.

The Recommender advised which UV it considered optimal as Plan A, and the other UV as Plan B, based on vehicle capabilities, capability weightings, and environmental factors that impacted capabilities. Automated advice was presented in the recommendation window (Fig. 2). Participants decided whether to agree with it and choose Plan A, or not to agree and choose Plan B. Participants had <60s to make this decision. The Recommender was 80% reliable, so it recommended the incorrect UV as Plan A on 20% of trials. In addition, on 10% of reliable trials, the Recommender advised the correct UV as Plan A, but made calculation error/s (see below). Feedback was provided following each trial specifying whether the recommendation was correct, the participants decision was correct, and their decision time.

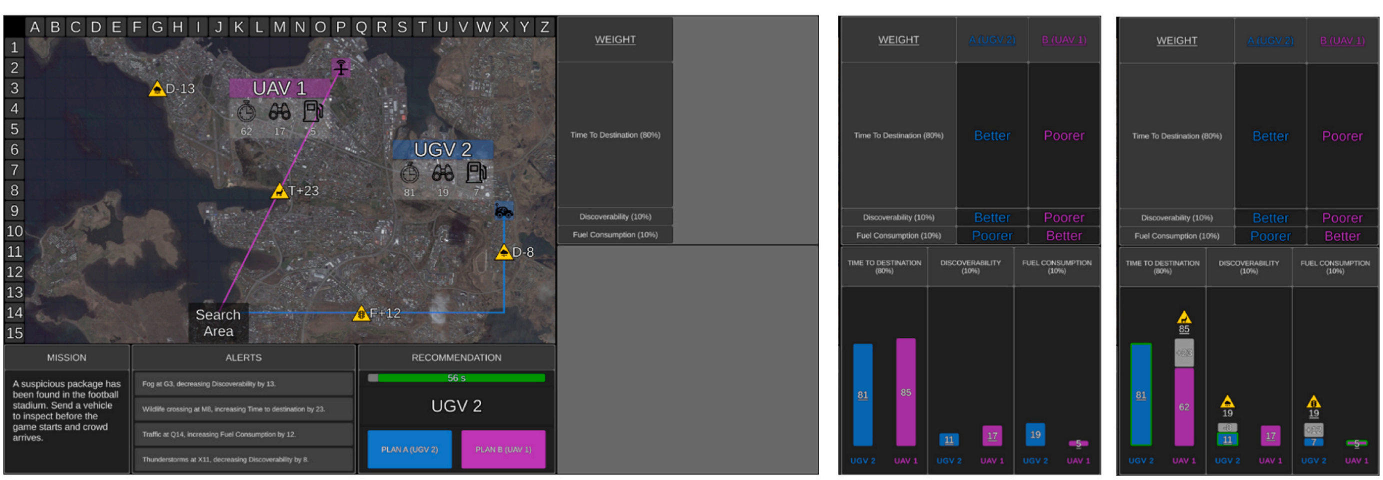
Transparency information was presented in the table and graph displays (Fig. 2). For all transparency conditions, the table showed how the Recommender evaluated the importance of each capability (i.e., weightings). Table rows were larger for capabilities with higher weightings. For medium and high transparency, the table also showed which UV the Recommender considered to be "better" and "poorer" on each UV capability.

The graph display was only presented for medium and high transparency conditions and included 3 bar graphs (one for each UV capability), allowing direct comparison between the UVs. For each graph, the Recommender's advice (Plan A) was presented on the left-hand side and Plan B was presented on the right-hand side. As lower scores on each capability are better, shorter bars on the graphs indicated better UV capability.

For medium transparency, the bar graphs presented the Recommender's score for each capability after taking environmental factors into consideration (Fig. 2b). In contrast, for high transparency, the bar graphs also showed *how* the Recommender calculated the score for each capability (Fig. 2c). Specifically, when an environmental factor was impacting a capability, the hazard symbol was placed on top of the relevant bar on the graph, and the bar included the original capability score (identical to tactical map), as well as the value added or subtracted from the original score (grey portion of the bar) due to the factor. The Recommender's calculated score was then underlined. Finally, the UV that the Recommender considered to be better on each capability was outlined green.

When the Recommender advised the incorrect UV (20% of trials) it either: (1) completely missed a relevant environmental factor, (2) miscalculated the impact of a relevant factor, or (3) a combination of errors 1 and 2 across different relevant factors.

For medium transparency, if the Recommender missed a relevant environmental factor, the original score would be incorrectly presented



(a) (b) (c)

Fig. 2. *An example mission trial with low (a), medium (b), and high (c) transparency.*
Note. The tactical map, mission, alerts, and recommendation windows are identical regardless of the transparency condition. The Recommender's advice can be seen in the recommendation window. In this example, the Recommender has recommended UGV 2 as Plan A and UAV 1 as Plan B. Transparency information is presented in the table (top right) and graph (bottom right; medium and high transparency conditions) displays. In both the table and graph displays, Plan A is presented on the left and Plan B on the right. The table display presents how the Recommender has evaluated the weighting of each capability, with larger rows signify higher weightings. It also outlines which UV the Recommender considered to be better and poorer on each capability (identical for medium (b) and high (c) transparency conditions). For medium transparency (b), the graph display presents the Recommender's calculated score for each capability, after taking environmental factors into consideration. Shorter bars indicate better performance. For high transparency (c), the graph display shows *how* the Recommender calculated the score for each capability. That is, when there is a relevant environmental factor, the bar included the original score as seen on the tactical map as well as the score that the Recommender added or subtracted from the original score which can be seen in the grey portion of the bar. The Recommender's calculated score was underlined. Also, the relevant environmental hazard symbol that the Recommender has factored in was presented on the top of the relevant bar. Finally, in high transparency the UV that the Recommender considered to be better on each capability was outlined green. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

instead of the new calculated score in the graph display. If the Recommender miscalculated the impact of a relevant environmental factor, the calculated score would be incorrect in the graph display.

For high transparency, if the Recommender missed a relevant environmental factor, the hazard symbol for that UV capability would be missing from the graph display to indicate it was not considered, and the original score would be incorrectly presented instead of the new calculated score. If the Recommender detected but miscalculated the impact of a relevant environmental factor, the hazard symbol for that UV capability would be displayed, but the score that the recommender added or subtracted would be incorrect, and thus the score would be incorrect. In addition, if the Recommender missed or miscalculated a relevant environmental factor, the green border could be placed around the incorrect UV's bar on the graph. Given such errors, in the table display, the UV that the Recommender considered to be "better" and "poorer" on each capability may also be incorrect.

On 10% of the reliable trials, the Recommender advised the correct UV, but it still made one of the three errors mentioned above. However, these errors were insignificant and did not result in the incorrect UV being recommended. These trials were included to minimize participants automatically selecting Plan B as soon as they detected any type of error made by the Recommender in the graph display.

The 100 trials were grouped into two 50-trial blocks. In both blocks there was a similar number of relevant environmental factors, capability weighting sets, environmental factor impact, and type of Recommender errors (Appendix A and B). The presentation order of blocks was counterbalanced, as well as whether the image analysis task was present or not for the first block completed. The order of trials presented within blocks was randomised but yoked across the transparency conditions.

2.2.2. Image analysis task

The image analysis task (Fig. 3) presented an aerial map with red square, yellow triangle, green circle, and blue hexagon icons. The number of icons presented varied from three to six. Participants identified when one of three changes occurred: the appearance, disappearance, or movement of an icon, by clicking on the relevant button (A = appear, D = disappear, M = move). Participants had 5s to respond. There were three changes per minute, with 6–15s between each. This task ran continuously across mission trials and no automated assistance was provided.

2.3. Measures

2.3.1. Automated task performance

Hit rate was the proportion of trials that the participant selected Plan A when it was correct. Correct rejection rate was the proportion of trials that the participant selected Plan B when the recommended Plan A was incorrect. The Signal Detection parameter d' assessed participants' sensitivity to correctly discriminate Plan A from B. The parameter c assessed participants bias towards agreeing with automation (negative values = greater bias). To adjust for extreme hit or false alarm values (0 or 1), rates of 0 were replaced with 0.5/n, whilst rates of 1 were replaced with (n – 0.5)/n, where n is the number of signal (Plan A correct) or noise (Plan B correct) trials (Macmillan and Kaplan, 1985). Decision times were based on correct decisions.

2.3.2. Concurrent task performance

A hit was recorded if participants responded to a change within 5s (regardless of the change type selected), otherwise a miss was recorded. Hit rate was calculated as the number of icon changes detected divided by the number of presented changes. A false alarm was recorded if participants clicked on one of the buttons when there was no change (i.e., >5s from a change). We analysed raw false alarms (as the denominator for a false alarm rate calculation was not determinable). Response times (RTs) were based on correct change identifications.

2.3.3. Workload

After each mission, subjective workload was measured using The Air Traffic Workload Input Technique (ATWIT; Stein, 1985). Participants had 5s to rate their workload from 1 (very low) to 10 (very high) and were instructed to consider the combined demands of both tasks under conditions of concurrent task demands. The image analysis task was paused whilst participants rated workload.

2.3.4. System usability scale

Perceived usability was measured using an adapted version of the 10-item System Usability Scale (SUS; Brooke, 1996). Scale reliability ranges from $\alpha = 0.83$ to $\alpha = 0.97$ (Lewis, 2018). An example item is "I thought the Recommender was easy to use". Items were rated on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). Some items were reverse-scored and then all items added and multiplied by 2.5 to yield a score from 0 to 100.

2.3.5. Trust questionnaire

Trust was measured using a six-item questionnaire adapted from

Fig. 3. Image analysis task display.
Note. The image analysis task display included an aerial view of a map with icons. Icons include red squares, yellow triangles, green circles, and blue hexagons. The display also includes three buttons (A = appear, D = disappear, M = move). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Merritt (2011). Scale reliability ranges from $\alpha = 0.87$ to $\alpha = 0.92$ (Merritt, 2011). An example item is "I believe the Recommender is a competent performer". Items were rated on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree).

2.4. Procedure

The experiment duration was 2hrs. Participants completed 60mins of training. Training began with a 20min audio-visual PowerPoint presentation explaining how to manually complete the mission and image analysis tasks. Participants then completed three blocks of practice trials without automation, including 10 trials of the mission task alone, 3mins of the image analysis task alone, then 10 trials of the two tasks performed concurrently. Participants then watched a 20min audio-visual PowerPoint presentation specific to their transparency condition, which explained the automation. Participants then completed 100 test trials. Participants took a break after 50 trials to complete the trust and SUS questionnaires (in a counterbalanced order). Participants completed these questionnaires again at the end of the experiment.

3. Results

Two participants were excluded (one medium, one high transparency) as they did not follow task instructions. Table 2 presents descriptive statistics for dependent variables during training trials (manual), split by transparency condition. We ran one-way analyses of variance (ANOVAs) to examine differences in performance (mission task, image analysis task) and workload ratings between transparency conditions during training. There were no significant differences on these outcome variables during training as a function of transparency (smallest $p = .14$), suggesting that our conditions were equivalent at baseline.

Table 3 presents descriptive statistics and correlations between dependent variables on test trials (collapsed across transparency and concurrent task conditions). Several expected correlations emerged. A strong positive correlation was found between trust and perceived usability. A moderate negative correlation was found between correct rejection rates and trust, and between decision time and trust. A moderate negative correlation was found between criterion (bias) and trust, and criterion and usability. A moderate positive correlation was found between criterion and decision time.

Table 4 presents descriptive statistics for dependent variables split by transparency and concurrent task demands. To test our predictions, we ran a series of 3 Transparency (low, medium, high) x 2 Concurrent Task (present, absent) mixed ANOVAs. Transparency was the between-subjects factor and concurrent task the within-subjects factor. Main effects of transparency, or interactions between transparency and concurrent task demands, were followed up with planned orthogonal contrasts that directly paralleled our hypotheses (Rosenthal and Rosnow, 1985) by comparing high to medium transparency, and high to low transparency, when the concurrent task was present and absent. High transparency served as the benchmark condition, and as such we did not compare the low and medium transparency conditions. This planned comparison approach minimized the potential problem of capitalizing on chance that could arise from statistically parcelling the same data in multiple ways (Rosenthal and Rosnow, 1985). To assess the impact of transparency on image analysis task accuracy and RT, we ran one-way ANOVAs. Significant one-way ANOVAs were followed up with planned orthogonal contrasts comparing high to medium transparency, and high to low transparency. Effect sizes were estimated using partial eta squared (small = .01, medium = 0.06, large = 0.14) for F-tests and Cohen's d (small = 0.20, medium = 0.50, large = 0.80; Cohen, 1992) for t-tests.

Table 2
Descriptive statistics for manual mission task performance, workload, and image analysis task performance during training, split by transparency condition.

	Mission Task		Workload	Image Analysis		
	Hit	DT	ATWIT	Hit	RT	FA
			Concurrent Task Absent			
Low	.89	18.63	4.01			
	(.10)	(7.34)	(1.33)			
Medium	.89	19.53	3.84			
	(.12)	(6.37)	(1.51)			
High	.86	19.05	3.83			
	(.13)	(6.11)	(1.36)			
			Concurrent Task Present			
Low	.79	20.18	4.92	.61	2.31	2.18
	(.14)	(6.76)	(1.82)	(.21)	(.48)	(1.68)
Medium	.78	20.06	4.87	.61	2.31	2.55
	(.14)	(5.65)	(1.90)	(.20)	(.55)	(2.05)
High	.78	20.72	4.91	.62	2.23	2.06
	(.18)	(7.38)	(1.56)	(.22)	(.38)	(2.06)

Note. There were 10 mission trials when the concurrent task was present and 10 mission trials when it was absent. Standard deviations are presented in parentheses. Decision time was measured in seconds. CR = correct rejection; DT = correct decision time; ATWIT = Air Traffic Workload Input Technique; FA = false alarm (raw number); RT = response time.

3.1. Accuracy of automation use

There was a main effect of transparency on hit rates, $F(1,207) = 10.79$, $p < .001$, $\eta_p^2 = .09$, but no main effect of concurrent task, $F < 1$, and no interaction, $F < 1$. Hit rates were higher with high than medium transparency both when the concurrent task was present, $t(137) = 2.78$, $p = .01$, $d = 0.43$, and absent, $t(137) = 2.82$, $p = .01$, $d = 0.53$. Hit rates were also higher with high than low transparency both when the concurrent task was present, $t(138) = 4.44$, $p < .001$, $d = 0.71$, and absent, $t(138) = 3.70$, $p < .001$, $d = 0.67$.

There was a main effect of concurrent task on correct rejection rates, $F(1,207) = 5.56$, $p = .02$, $\eta_p^2 = .03$, indicating that correct rejections decreased when the concurrent task was present ($M = 0.67$, $SD = 0.27$) compared to absent ($M = 0.71$, $SD = 0.25$). There was no main effect of transparency, $F < 1$, and no interaction, $F(1,207) = 1.41$, $p = .25$, $\eta_p^2 = .01$.

There was a main effect of concurrent task on sensitivity (d'), $F(1,207) = 9.76$, $p = .002$, $\eta_p^2 = .05$, with participants less sensitive (i.e., to correctly discriminate Plan A from B) when the concurrent task was present ($M = 1.84$, $SD = 0.99$) compared to absent ($M = 2.02$, $SD = 1.01$). There was a main effect of transparency, $F(1,207) = 4.05$, $p = .02$, $\eta_p^2 = .04$, but no interaction, $F(1,207) = 1.77$, $p = .17$, $\eta_p^2 = .02$. Participants were more sensitive with high than low transparency both when the concurrent task was present, $t(138) = 2.50$, $p = .01$, $d = .42$, and absent, $t(138) = 2.58$, $p = .01$, $d = 0.43$. However, there was no difference in sensitivity between high and medium transparency conditions when the concurrent task was present, $t < 1$, or absent, $t(137) = 1.88$, $p = .06$, $d = 0.32$.

For response bias (c) there was a main effect of transparency, $F(1,207) = 3.46$, $p = .03$, $\eta_p^2 = .03$, but no main effect of concurrent task, $F(1,207) = 2.24$, $p = .14$, $\eta_p^2 = .01$, and no interaction, $F < 1$. Participants were more biased towards agreeing with automation with high than low transparency when the concurrent task was absent, $t(138) = 2.13$, $p = .04$, $d = .37$, but not when it was present, $t(138) = 1.62$, $p = .11$, $d = 0.28$. In contrast, participants were more biased towards agreeing with automation with high than medium transparency when the concurrent task was present, $t(137) = 2.62$, $p = .01$, $d = 0.45$, but not when absent $t(137) = 1.57$, $p = .12$, $d = 0.26$.

3.2. Correct decision time

There was a main effect of concurrent task on correct decision time, F

Table 3
Descriptive statistics and inter-correlations between accuracy of automation use, decision time, perceived workload, trust, and usability, collapsed across transparency and concurrent task conditions.

Variable	*M*	SD	1	2	3	4	5	6	7
1. Hit	.91	.07							
2. CR	.69	.23	.27**						
3. DT	15.55	5.35	−.27**	.25**					
4. *d'*	1.93	.92	.65**	.89**	.04				
5. *c*	−.43	.36	−.29**	.82**	.37**	.49**			
6. Workload	4.75	1.58	−.11	.03	.24**	−.04	.09		
7. Trust	2.72	.82	−.03	−.38**	−.30**	−.29**	−.35**	−.07	
8. Usability	61.45	14.92	.19**	−.17*	−.29**	−.04	−.30**	−.21**	.61**

Note. Decision time was measured in seconds. CR = correct rejection; DT = correct decision time; *d'* = sensitivity; *c* = criterion. Weak, moderate, and strong effect sizes are 0.10, 0.30, and 0.50, respectively (Cohen, 1992) *$p < .05$. **$p < .01$.

Table 4
Descriptive statistics for accuracy of automation use, correct decision time, workload, trust, usability, and image analysis task performance split by transparency condition and whether the concurrent task was present or absent.

	Automation Use					Workload	Trust	Usability	Image Analysis		
Transparency	Hit	CR	DT	*d'*	*c*	ATWIT	Rating	Rating	Hit	RT	FA
					Concurrent Task Absent						
Low	.89 (.08)	.70 (.25)	13.22 (5.30)	1.84 (.91)	−.35 (.41)	4.41 (1.34)	2.84 (.84)	62.85 (14.57)			
Medium	.90 (.08)	.70 (.26)	14.09 (5.60)	1.94 (1.01)	−.39 (.43)	4.13 (1.55)	2.58 (.94)	59.09 (17.00)			
High	.94 (.07)	.72 (.25)	12.61 (5.09)	2.27 (1.08)	−.49 (.35)	4.24 (1.53)	2.80 (.86)	64.28 (16.79)			
					Concurrent Task Present						
Low	.88 (.07)	.64 (.28)	18.05 (6.51)	1.59 (.89)	−.43 (.49)	5.31 (1.91)	2.76 (.91)	61.94 (14.19)	.63 (.17)	2.25 (.35)	9.20 (6.51)
Medium	.90 (.07)	.70 (.23)	19.21 (6.43)	1.91 (.87)	−.38 (.35)	5.22 (2.06)	2.46 (.81)	58.91 (14.79)	.58 (.15)	2.30 (.28)	12.37 (7.52)
High	.93 (.07)	.66 (.30)	16.07 (7.11)	2.02 (1.15)	−.56 (.44)	5.19 (1.69)	2.86 (.82)	61.49 (16.48)	.60 (.19)	2.23 (.30)	8.33 (7.21)

Note. Standard deviations are presented in parentheses. Decision time was measured in seconds. CR = correct rejection; DT = correct decision time; *d'* = sensitivity; *c* = criterion; ATWIT = Air Traffic Workload Input Technique; FA = false alarm (raw number); RT = response time.

(1,207) = 122.01, $p < .001$, $\eta_p^2 = .37$, with decision times slower when the concurrent task was present ($M = 17.78$s, $SD = 6.78$), compared to absent ($M = 13.31$s, $SD = 5.34$). There was a main effect of transparency, $F(1,207) = 3.33$, $p = .04$, $\eta_p^2 = .03$, but no interaction, $F(1,207) = 1.57$, $p = .21$, $\eta_p^2 = .02$. Decision times were faster with high than medium transparency when the current task was present, $t(137) = 2.73$, $p = .01$, $d = 0.46$, but not when absent, $t(137) = 1.64$, $p = .10$, $d = 0.28$. However, there was no difference in decision times between high and low transparency conditions when the current task was present, $t(138) = 1.72$, $p = .09$, $d = 0.29$, or absent, $t < 1$.

3.3. Workload

There was a main effect of concurrent task, $F(1,207) = 100.49$, $p < .001$, $\eta_p^2 = .39$, with perceived workload higher when the concurrent task was present ($M = 5.24$, $SD = 1.88$) compared to absent ($M = 4.26$, SD = 1.47). There was no main effect of transparency, $F < 1$, and no interaction, $F < 1$.

3.4. Trust and usability

For trust, there was a main effect of transparency, $F(1,205) = 3.13$, $p = .05$, $\eta_p^2 = .03$, but no main effect of concurrent task, $F < 1$, and no interaction, $F(1,205) = 2.37$, $p = .10$, $\eta_p^2 = .02$. Trust ratings were higher with high than medium transparency when the concurrent task was present, $t(137) = 2.91$, $p = .004$, $d = 0.49$, but not when absent, $t(137) = 1.40$, $p = .16$, $d = 0.24$. There were no differences in trust ratings between high and low transparency conditions when the concurrent task was present, $t < 1$, or absent, $t < 1$.

For usability, there was no main effect of concurrent task, $F(1,205) = 3.57$, $p = .06$, $\eta_p^2 = .02$, or transparency, $F(1,205) = 1.38$, $p = .26$, $\eta_p^2 = .01$, and no interaction, $F(1, 205) = 1.22$, $p = .30$, $\eta_p^2 = .01$.

3.5. Image analysis task

There was no difference in image analysis task hit rates, $F(1,205) = 1.12$, $p = .33$, $\eta^2 = .01$, or RTs, $F < 1$, as a function of transparency. However, there was a significant difference in false alarms between transparency conditions, $F(2,209) = 6.28$, $p = .002$, $\eta^2 = .06$. Less false alarms were made with high than medium transparency, $t(137) = 3.23$, $p = .002$, $d = 0.55$, with no difference in false alarms between high and low transparency conditions, $t < 1$.

4. Discussion

We examined the impact of increased transparency on the accuracy of automation use, decision time, subjective workload, and perceived trust/usability in a UV task, under single and concurrent task (dual-task) conditions. Low transparency presented information regarding how the Recommender evaluated the importance of each UV capability based on weightings. In addition, medium transparency provided a comparison between which UV was better and poorer on each capability, as well as a visual comparison of calculated UV scores. High transparency was the benchmark condition which additionally provided a visualization of information regarding *how* the Recommender calculated scores.

4.1. Impact of transparency without concurrent task demands

When concurrent task demands were absent, despite a bias towards agreeing with automated advice, high transparency resulted in more accurate automation use (sensitivity to correctly discriminate Plan A from B) compared to low transparency. While not reaching significance, there was a trend ($p = .06$) towards more accurate automation use with high compared to medium transparency. There were no costs to decision times or perceived workload, and no differences in trust or usability ratings as a function of transparency.

These findings that increased transparency benefited the accuracy of automation use, without costs to perceived workload or decision time, are generally in line with prior work in UV task settings (e.g., Mercado et al., 2016), as well as other task domains (see Van de Merwe et al., 2022). Contrary to the findings of Stowers et al. (2020) we did not find increased decision time with increased transparency. A key difference between the two studies is that our transparency was provided in visual icons/numbers, whereas Stowers et al. used visual and text explanations, potentially increasing the time taken to process increased transparency information. Contrary to some previous findings (see Bhaskara et al., 2020 for review), increased transparency did not improve perceived trust or usability.

4.2. Impact of transparency with concurrent task demands

In line with prior research (e.g., Bagheri and Jamieson, 2004; Bailey and Scerbo, 2007), the accuracy of automation use (sensitivity) decreased (including a significant reduction in correct rejections), decision times slowed, and workload was rated higher, with concurrent task demands. Despite this, when the concurrent task was present, the benefits of high transparency compared to low transparency for the accuracy of automation use (sensitivity) remained significant (and with an almost identical effect size to when the concurrent task was absent), with no bias towards agreeing with the automation, and no high transparency associated costs to decision time or workload. There was no difference in image analysis task performance for the high compared to low transparency condition, and thus no evidence of any differential dual task trade-offs as a function of high/low transparency.

Comparing the high and medium transparency conditions, when the concurrent task was present participants provided high transparency trusted the automation more, made faster decisions, and were more biased towards agreeing with the automation (as indicated by the higher hit rates and lower correct rejection rates compared to medium transparency). Taken together, these findings indicate increased reliance on automation when high compared to medium transparency was provided under conditions in which participants performed a concurrent non-automated task, potentially associated with an automation monitoring strategy in which reduced attention was allocated to process the high transparency information to verify automated advice (Bhaskara et al., 2021; Moray, 1986; Wickens et al., 2015). In line with this conclusion, participants provided high transparency made less false alarms than those provided medium transparency on the image analysis task, potentially indicative that participants provided high transparency were paying more attention to the concurrent image analysis task than those provided medium transparency. These points notwithstanding, future research that employs eye tracking, or some form of transparency information unmasking procedures, will be required to precisely (objectively) examine the relative attention paid to the mission task and image analysis task as a function of transparency condition.

Although there was no difference in the accuracy of automation use (sensitivity) with high compared to medium transparency with concurrent task demands, the increased trust ratings and automation bias have potentially important practical implications. In operational environments with particularly highly reliable automation (e.g., 95% reliable) and concurrent task demands, high transparency may lead to pronounced misuse due to lowered expectations of incorrect automated advice and associated over-reliance on automation. A potential design solution to mitigate this is to design adaptive automation which directs operator attention away from the concurrent task and towards automation transparency information when it is most likely necessary (e.g., when there is a marked decline in automation verification rates and/or performance) (Scerbo, 1996; Vagia et al., 2016).

Furthermore, under conditions in which concurrent task demands have a high information processing load, there could be increased potential for operators provided high transparency to further rely on automated advice and thus not allocate the attentional capacity required to scrutinise the high transparency information provided in order to verify automated advice. A real-world instance of where this could be problematic is in robotic swarm management, where operators are controlling or supervising multiple UVs and switching between many tasks (Hocraffer and Nam, 2017; Hussein et al., 2020). A potential design solution may be to adapt the presentation of automation transparency information based on the ongoing types and complexities of concurrent task demands. However, this design solution still requires an empirical evidence-base for appropriate guidance.

4.3. Limitations and conclusion

This UV task is broadly representative of other task contexts in which operators rely on automation to assist with decisions when they have concurrent task demands. That said, there are potential problems in generalising from novice participants to experts as there are undoubtedly differences in their cognitive skills and motivation (Rieth and Hagemann, 2022). Future research should re-examine these research questions with expert UV operators, although it can be challenging to find enough experts for required statistical power.

In conclusion, to our knowledge this is the first study to demonstrate that high transparency can be designed to improve the accuracy of automation use compared to low transparency under conditions with concurrent task demands. Outcomes when comparing high to medium transparency were more nuanced. There was a trend toward high transparency improving the accuracy of automation use compared to medium transparency with no concurrent task demands, but with concurrent task demands there were clear indications of increased reliance on highly transparent automation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

A. List of Mission Descriptions and Trial Characteristics for Set A

Trial ID	Map Area	Mission Statements	Weighting	Relevant Environmental Factors	Overall Impact of Relevant Environmental Factors	Type of Error	Correct Plan
1	Urban	A large protest has erupted around Parliament House. Provide reconnaissance as events unfold.	40, 40, 20	2	Positive	N/A	A
2	Coastal	Illegal fishing has been reported. Patrol the area and intercept any vessels detected.	40, 40, 20	3	Positive	N/A	A
3	Rural	Five people have not been accounted for in a bushfire. Send a vehicle to search for these people.	40, 40, 20	1	Positive	N/A	A

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Trial ID	Map Area	Mission Statements	Weighting	Relevant Environmental Factors	Overall Impact of Relevant Environmental Factors	Type of Error	Correct Plan
4	Urban	A police chase is underway on the freeway. Send a vehicle to assist.	45, 45, 10	3	Positive	N/A	A
5	Coastal	A large shipment of illegal drugs has been spotted in pirate-infested waters. Send a vehicle to identify and intercept the ship.	45, 45, 10	1	Positive	N/A	A
6	Rural	A pilot made a mayday call before crashing and losing contact. Send a vehicle to find the location of the crash site.	45, 45, 10	2	Positive	N/A	A
7	Urban	There is traffic congestion along the freeway. Send a vehicle to monitor the flow of traffic.	40, 30, 30	2	Positive	N/A	A
8	Coastal	Provide security and surveillance support for a fleet of cargo ships as they traverse pirate-infested waters.	40, 30, 30	3	Positive	N/A	A
9	Rural	There has been a traffic accident. Send a vehicle to assess the situation.	40, 30, 30	1	Positive	N/A	A
10	Urban	An unidentified UAV is spotted flying close to homes and hacking into Wi-Fis. Send a vehicle to search, identify, and intercept.	60, 20, 20	3	Positive	N/A	A
11	Coastal	A distress call was received from a boat due to engine failure. Send a vehicle to search for the exact location of the boat.	60, 20, 20	2	Positive	N/A	A
12	Rural	Demolitions to create a new open-cut mine are due to commence. Monitor the area and report any vehicles or people detected.	80, 10, 10	1	Positive	N/A	A
13	Coastal	A pilot made a mayday call off the coast before losing contact. Send a vehicle to check whether the aircraft has crashed and if there were any survivors.	80, 10, 10	2	Positive	N/A	A
14	Urban	A group of teenagers have been seen breaking into and vandalising warehouses. Send a vehicle to the area to confirm this.	40, 40, 20	1	Positive	Missed	B
15	Coastal	An aircraft has lost contact off the coast. Find it as soon as possible.	45, 45, 10	3	Positive	Missed (1) Miscalculated (2)	B
16	Rural	Poachers have been reported to be hunting within a protected national park. Send a vehicle to intercept them.	40, 30, 30	2	Positive	Missed (1) Correct (1)	B
17	Urban	A suspicious boat is about to dock on the beach near the prison. Send an air vehicle to provide surveillance of the area.	60, 20, 20	2	Positive	Miscalculated (1) Correct (1)	B
18	Coastal	Empty rocket boosters have detached from a spaceship and are expected to land off the coast. Monitor the area to track its landing location.	80, 10, 10	1	Positive	Missed	B
19	Rural	Reports of people illegally dumping rubbish on the beach. Send a vehicle to deter litterers.	60, 20, 20	1	Positive	Miscalculated	A
20	Urban	Escort Air Force One as it returns to the US. Maintain surveillance of the surrounding airspace.	60, 20, 20	1	Negative	N/A	A
21	Coastal	Unidentified cargo ship is planning to dock on the island's coast. Send a vehicle to inspect and identify the ship.	60, 20, 20	3	Negative	N/A	A
22	Rural	A bushfire is burning out of control in the mountain ranges. Send a vehicle to search for hikers trapped in the area.	60, 20, 20	2	Negative	N/A	A
23	Urban	A suspicious package has been found in the football stadium. Send a vehicle to inspect before the game starts and crowd arrives.	80, 10, 10	3	Negative	N/A	A
24	Coastal	A suspicious boat is operating aimlessly around the Islands. Provide surveillance for a potential attack.	80, 10, 10	2	Negative	N/A	A
25	Rural	An earthquake caused structural damage to some houses. Send a vehicle to identify the damage.	80, 10, 10	1	Negative	N/A	A
26	Urban	Buildings have collapsed due to an earthquake. Search for survivors.	40, 40, 20	1	Negative	N/A	A
27	Coastal	A distress call has been received from a ship off the coast. Send a vehicle to monitor and assist rescue operations.	40, 40, 20	2	Negative	N/A	A
28	Rural	A vehicle has been detected moving at high speed directly towards a Defence Base. Intercept the vehicle before it reaches the perimeter.	40, 40, 20	3	Negative	N/A	A
29	Urban	An armed fugitive is hiding in an abandoned building. Send a vehicle to provide surveillance for police.	45, 45, 10	2	Negative	N/A	A
30	Coastal	Swimming marathon is on and a shark has been sighted nearby. Monitor the area for the duration of the swimming event.	45, 45, 10	1	Negative	N/A	A
31	Rural	The southern region is holding its annual health fair. Send a vehicle to deliver medicine and other medical supplies.	40, 30, 30	3	Negative	N/A	A
32	Urban	An unidentified aircraft has been detected flying around a prison. Search for and intercept the aircraft as soon as possible.	40, 30, 30	3	Negative	N/A	A
33	Urban	Violence has erupted at the football stadium. Send a vehicle to provide support to police.	40, 40, 20	2	Negative	Missed (1) Miscalculated (1)	B

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Trial ID	Map Area	Mission Statements	Weighting	Relevant Environmental Factors	Overall Impact of Relevant Environmental Factors	Type of Error	Correct Plan
34	Coastal	Suspicious submerged object has been reported in the harbour. Send a vehicle to identify.	45, 45, 10	1	Negative	Missed	B
35	Rural	A report is received about a drone spying on homes. Send a vehicle to find and track it.	40, 30, 30	3	Negative	Missed (1) Correct (2)	B
36	Coastal	A cruise ship has been quarantined due to a virus outbreak. Send a vehicle to deliver medical supplies.	60, 20, 20	2	Negative	Missed (1) Correct (1)	B
37	Rural	Heavy rains have caused rivers from the South-eastern mountains to flood. Send a vehicle to monitor the flood waters.	80, 10, 10	3	Negative	Miscalculated (2) Correct (1)	B
38	Urban	The International Air show is on. Maintain surveillance of the surrounding airspace to ensure non-authorised aircraft do not enter.	40, 30, 30	2	Negative	Miscalculated (2)	A
39	Urban	A high value prisoner is being transferred between prisons. Provide surveillance over their route as they leave the city.	80, 10, 10	3	Positive	N/A	A
40	Coastal	A giant oil rig is leaking oil into the ocean. Monitor the area to track the oil as it spreads.	60, 20, 20	1	Positive	N/A	A
41	Rural	A pilot made a mayday call around the mountain ranges before losing contact. Send a vehicle to check whether the aircraft crashed and if there were any survivors.	40, 30, 30	2	Positive	N/A	A
42	Coastal	A ferry is reported sinking after colliding with a cargo ship. Send a vehicle to find these people.	45, 45, 10	1	Positive	Miscalculated	A
43	Urban	There is a police chase on the freeway. Send a vehicle to monitor the chase and intercept.	40, 30, 30	1	Negative	N/A	A
44	Coastal	A shark has been spotted off the coast. Send a vehicle to patrol the area.	45, 45, 10	3	Negative	N/A	A
45	Rural	Crops have been damaged on farms in the northern region. Send a vehicle to assess the extent of the damage.	80, 10, 10	2	Negative	N/A	A
46	Urban	An unidentified UAV is being used to spray paint and vandalise billboards along the freeway. Send a vehicle to intercept.	80, 10, 10	3	Negative	Missed (2) Miscalculated (1)	A
47	Urban	Police are searching for a stolen car being driven around the northern business district. Send a vehicle to search the area.	40, 40, 20	0	Irrelevant	N/A	A
48	Coastal	A pilot made a mayday call above pirate-infested water, before crashing and losing contact. Send a vehicle to check the location of the aircraft and if there were any survivors.	45, 45, 10	0	Irrelevant	N/A	A
49	Rural	There have been reports of stolen cattle on farms. Send a vehicle to monitor the area.	60, 20, 20	0	Irrelevant	N/A	A
50	Rural	There is a fire in the southern bushlands. Send a vehicle to map hotspots and high-risk areas.	40, 20, 20	3	Positive	Missed (2) Miscalculated (1)	A

Appendix B

A List of Mission Descriptions and Trial Characteristics for Set B

Trial ID	Map Area	Mission Statements	Weighting	Relevant Environmental Factors	Overall Impact of Relevant Environmental Factors	Type of Error	Correct Plan
51	Urban	An unidentified UAV is being used to spray paint and vandalise billboards along the freeway. Send a vehicle to intercept	60, 20, 20	2	Positive	N/A	A
52	Coastal	A pilot made a mayday call above pirate-infested water, before crashing and losing contact. Send a vehicle to check the location of the aircraft and if there were any survivors.	60, 20, 20	1	Positive	N/A	A
53	Rural	A bushfire is burning out of control in the mountain ranges. Send a vehicle to search for hikers trapped in the area.	60, 20, 20	3	Positive	N/A	A
54	Urban	Police are searching for a stolen car being driven around the northern business district. Send a vehicle to search the area.	80, 10, 10	2	Positive	N/A	A
55	Coastal	Unidentified cargo ship is planning to dock on the island's coast. Send a vehicle to inspect and identify the ship.	80, 10, 10	1	Positive	N/A	A
56	Rural	A pilot made a mayday call around the mountain ranges before losing contact. Send a vehicle to check whether the aircraft crashed and if there were any survivors.	80, 10, 10	3	Positive	N/A	A
57	Urban	A suspicious package has been found in the football stadium. Send a vehicle to inspect before the game starts and crowd arrives.	40, 40, 20	3	Positive	N/A	A
58	Coastal	A suspicious boat is operating aimlessly around the Islands. Provide surveillance for a potential attack.	40, 40, 20	2	Positive	N/A	A
59	Rural	An earthquake caused structural damage to some houses. Send a vehicle to identify the damage.	40, 40, 20	1	Positive	N/A	A

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Trial ID	Map Area	Mission Statements	Weighting	Relevant Environmental Factors	Overall Impact of Relevant Environmental Factors	Type of Error	Correct Plan
60	Urban	Buildings have collapsed due to an earthquake. Search for survivors.	45, 45, 10	1	Positive	N/A	A
61	Coastal	A distress call has been received from a ship off the coast. Send a vehicle to monitor and assist rescue operations.	45, 45, 10	2	Positive	N/A	A
62	Rural	A vehicle has been detected moving at high speed directly towards a Defence Base. Intercept the vehicle before it reaches the perimeter.	40, 30, 30	1	Positive	N/A	A
63	Urban	An armed fugitive is hiding in an abandoned building. Send a vehicle to provide surveillance for police.	40, 30, 30	3	Positive	N/A	A
64	Urban	A ferry is reported sinking after colliding with a cargo ship. Send a vehicle to find these people.	40, 40, 20	1	Positive	Miscalculated	B
65	Coastal	Swimming marathon is on and a shark has been sighted nearby. Monitor the area for the duration of the swimming event.	45, 45, 10	2	Positive	Missed (1) Correct (1)	B
66	Rural	There is a fire in the southern bushlands. Send a vehicle to map hotspots and high-risk areas.	40, 30, 30	3	Positive	Missed (3)	B
67	Urban	A high value prisoner is being transferred between prisons. Provide surveillance over their route as they leave the city.	60, 20, 20	2	Positive	Miscalculated (1) Correct (1)	B
68	Coastal	A giant oil rig is leaking oil into the ocean. Monitor the area to track the oil as it spreads.	80, 10, 10	3	Positive	Miscalculated (1) Correct (2)	B
69	Coastal	A ferry is reported sinking after colliding with a cargo ship. Send a vehicle to find these people.	40, 40, 20	3	Positive	Miscalculated (3)	A
70	Urban	The International Air show is on. Maintain surveillance of the surrounding airspace to ensure non-authorised aircraft do not enter.	40, 40, 20	2	Negative	N/A	A
71	Coastal	An aircraft has lost contact off the coast. Find it as soon as possible.	40, 40, 20	1	Negative	N/A	A
72	Rural	Heavy rains have caused rivers from the South-eastern mountains to flood. Send a vehicle to monitor the flood waters.	40, 40, 20	3	Negative	N/A	A
73	Urban	Escort Air Force One as it returns to the US. Maintain surveillance of the surrounding airspace.	45, 45, 10	2	Negative	N/A	A
74	Coastal	Empty rocket boosters have detached from a spaceship and are expected to land off the coast. Monitor the area to track its landing location.	45, 45, 10	1	Negative	N/A	A
75	Rural	Crops have been damaged on farms in the northern region. Send a vehicle to assess the extent of the damage.	45, 45, 10	3	Negative	N/A	A
76	Urban	Violence has erupted at the football stadium. Send a vehicle to provide support to police.	40, 30, 30	2	Negative	N/A	A
77	Coastal	Suspicious submerged object has been reported in the harbour. Send a vehicle to identify.	40, 30, 30	3	Negative	N/A	A
78	Rural	There have been reports of stolen cattle on farms. Send a vehicle to monitor the area.	40, 30, 30	1	Negative	N/A	A
79	Urban	There is a police chase on the freeway. Send a vehicle to monitor the chase and intercept.	60, 20, 20	1	Negative	N/A	A
80	Coastal	A cruise ship has been quarantined due to a virus outbreak. Send a vehicle to deliver medical supplies.	60, 20, 20	2	Negative	N/A	A
81	Rural	There has been a traffic accident. Send a vehicle to assess the situation.	80, 10, 10	3	Negative	N/A	A
82	Rural	Demolitions to create a new open-cut mine are due to commence. Monitor the area and report any vehicles or people detected.	80, 10, 10	1	Negative	N/A	A
83	Urban	A group of teenagers have been seen breaking into and vandalising warehouses. Send a vehicle to the area to confirm this.	40, 40, 20	3	Negative	Missed (2) Miscalculated (1)	B
84	Coastal	A shark has been spotted off the coast. Send a vehicle to patrol the area.	45, 45, 10	1	Negative	Miscalculated	B
85	Rural	Poachers have been reported to be hunting within a protected national park. Send a vehicle to intercept them.	40, 30, 30	2	Negative	Missed (1) Miscalculated (1)	B
86	Urban	A suspicious boat is about to dock on the beach near the prison. Send an air vehicle to provide surveillance of the area.	60, 20, 20	1	Negative	Missed	B
87	Coastal	A ferry is reported sinking after colliding with a cargo ship. Send a vehicle to find these people.	80, 10, 10	3	Negative	Missed (1) Correct (2)	B
88	Rural	A pilot made a mayday call before crashing and losing contact. Send a vehicle to find the location of the crash site.	80, 10, 10	2	Negative	Missed (1) Miscalculated (1)	A
89	Urban	An unidentified UAV is spotted flying close to homes and hacking into Wi-Fis. Send a vehicle to search, identify, and intercept.	60, 20, 20	1	Positive	N/A	A

(*continued on next page*)

(*continued*)

Trial ID	Map Area	Mission Statements	Weighting	Relevant Environmental Factors	Overall Impact of Relevant Environmental Factors	Type of Error	Correct Plan
90	Coastal	A pilot made a mayday call off the coast before losing contact. Send a vehicle to check whether the aircraft has crashed and if there were any survivors.	80, 10, 10	2	Positive	N/A	A
91	Rural	A report is received about a drone spying on homes. Send a vehicle to find and track it.	40, 40, 20	2	Positive	N/A	A
92	Urban	A police chase is underway on the freeway. Send a vehicle to assist.	60, 20, 20	3	Positive	Missed (2) Correct (1)	A
93	Urban	A large protest has erupted around Parliament House. Provide reconnaissance as events unfold.	45, 45, 10	3	Negative	N/A	A
94	Coastal	Illegal fishing has been reported. Patrol the area and intercept any vessels detected.	40, 30, 30	2	Negative	N/A	A
95	Rural	The southern region is holding its annual health fair. Send a vehicle to deliver medicine and other medical supplies.	80, 10, 10	3	Negative	N/A	A
96	Coastal	Provide security and surveillance support for a fleet of cargo ships as they traverse pirate-infested waters.	40, 30, 30	1	Negative	Miscalculated	A
97	Urban	There is traffic congestion along the freeway. Send a vehicle to monitor the flow of traffic.	45, 45, 10	0	Irrelevant	N/A	A
98	Coastal	A distress call was received from a boat due to engine failure. Send a vehicle to search for the exact location of the boat.	40, 30, 30	0	Irrelevant	N/A	A
99	Rural	Five people have not been accounted for in a bushfire. Send a vehicle to search for these people.	60, 20, 20	0	Irrelevant	N/A	A
100	Rural	Reports of people illegally dumping rubbish on the beach. Send a vehicle to deter litterers.	45, 45, 10	2	Positive	Missed (1) Miscalculated (1)	A

References

Abich IV, J., Reinerman-Jones, L., Matthews, G., 2017. Impact of three task demand factors on simulated unmanned system intelligence, surveillance, and reconnaissance operations. Ergonomics 60, 791–809. https://doi.org/10.1080/00140139.2016.1216171.

Bagheri, N., Jamieson, G.A., 2004. A sampling model to ascertain automation-induced complacency in multi-task environments. In: Human Error, Safety and Systems Development. Springer, Boston, MA, pp. 131–145. https://doi.org/10.1007/1-4020-8153-7_9.

Bailey, N.R., Scerbo, M.W., 2007. Automation-induced complacency for monitoring highly reliable systems: the role of task complexity, system experience, and operator trust. Theor. Issues Ergon. Sci. 8, 321–348. https://doi.org/10.1080/14639220500535301.

Bhaskara, A., Duong, L., Brooks, J., Li, R., McInerney, R., Skinner, M., et al., 2021. Effect of automation transparency in the management of multiple unmanned vehicles. Appl. Ergon. 90, 103243 https://doi.org/10.1016/j.apergo.2020.103243.

Bhaskara, A., Skinner, M., Loft, S., 2020. Agent transparency: a review of current theory and evidence. IEEE Transact. Human-Mach. Syst. 50, 215–224. https://doi.org/10.1109/THMS.2020.2965529.

Brooke, J., 1996. Sus: a "quick and dirty'usability. In: Usability Evaluation in Industry, vol. 189. https://doi.org/10.1201/9781498710411-35.

Bussone, A., Stumph, S., O'Sullivan, D., 2015. The role of explanations on trust and reliance in clinical decision support systems. IEEE Int. Conf. Healthc. Informat. 21, 160–169. https://doi.org/10.1109/ICHI.2015.26.

Calhoun, G.L., Ruff, H.A., Behymer, K.J., Frost, E.M., 2018. Human-autonomy teaming interface design considerations for multi-unmanned vehicle control. Theor. Issues Ergon. Sci. 19, 321–352. https://doi.org/10.1080/1463922X.2017.1315751.

Chen, J.Y., Procci, K., Boyce, M., Wright, J., Garcia, A., Barnes, M., 2014. *Situation Awareness-Based Agent Transparency* (ARL-TR-6905). U.S. Army Research Laboratory. https://doi.org/10.21236/ada600351.

Christoffersen, K., Woods, D.D., 2002. 1. How to make automated systems team players. In: Advances in Human Performance and Cognitive Engineering Research. Emerald Group Publishing Limited. https://doi.org/10.1016/S1479-3601(02)02003-9.

Cohen, J., 1992. Statistical power analysis. Curr. Dir. Psychol. Sci. 1, 98–101.

Dorneich, M.C., Dudley, R., Letsu-Dake, E., Rogers, W., Whitlow, S.D., Dillard, M.C., Nelson, E., 2017. Interaction of automation visibility and information quality in flight deck information automation. IEEE Transact. Human-Mach. Syst. 47, 915–926. https://doi.org/10.1109/THMS.2017.2717939.

Dzindolet, M.T., Peterson, S.A., Pomranky, R.A., Pierce, L.G., Beck, H.P., 2003. The role of trust in automation reliance. Int. J. Hum. Comput. Stud. 58, 697–718. https://doi.org/10.1016/S1071-5819(03)00038-7.

Fu, W.T., Gray, W.D., 2006. Suboptimal tradeoffs in information seeking. Cognit. Psychol. 52, 195–242. https://doi.org/10.1016/j.cogpsych.2005.08.002.

Guznov, S., Lyons, J., Pfahler, M., Heironimus, A., Woolley, M., Friedman, J., Neimeier, A., 2020. Robot transparency and team orientation effects on human–robot teaming. Int. J. Hum. Comput. Interact. 36, 650–660. https://doi.org/10.1080/10447318.2019.1676519.

Hocraffer, A., Nam, C.S., 2017. A meta-analysis of human-system interfaces in unmanned aerial vehicle (UAV) swarm management. Appl. Ergon. 58, 66–80. https://doi.org/10.1016/j.apergo.2016.05.011.

Hussein, A., Elsawah, S., Abbass, H.A., 2020. Trust mediating reliability-reliance relationship in supervisory control of human-swarm interactions. Hum. Factors 62, 1237–1248. https://doi.org/10.1177/0018720819879273.

Kaber, D.B., 2018. Issues in human–automation interaction modelling: presumptive aspects of frameworks of types and levels of automation. J. Cogn. Eng. Decis. Mak. 12, 7–24. https://doi.org/10.1177/1555343417737203.

Kahneman, D., 1973. Attention and Effort. Prentice Hall, Englewood Cliffs, NJ.

Karpinsky, N.D., Chancey, E.T., Palmer, D.B., Yamani, Y., 2018. Automation trust and attention allocation in multitasking workspace. Appl. Ergon. 70, 194–201. https://doi.org/10.1016/j.apergo.2018.03.008.

Lee, J.D., See, K.A., 2004. Trust in automation: designing for appropriate reliance. Hum. Factors 46, 50–80. https://doi.org/10.1518/hfes.46.1.50_30392.

Lewis, J.R., 2018. The system usability scale: past, present, and future. Int. J. Hum. Comput. Interact. 34, 577–590. https://doi.org/10.1080/10447318.2018.1455307.

Loft, S., Bhaskara, A., Lock, B.A., Skinner, M., Brooks, J., Li, R., Bell, J., 2022. The Impact of Transparency and Decision Risk on Human–Automation Teaming Outcomes. Human Factors, 00187208211033445. https://doi.org/10.1177/00187208211033445.

Lyons, J.B., 2013. Being transparent about transparency: a model for human-robot interaction. In: 2013 AAAI Spring Symposium Series.

Lyons, J.B., Stokes, C.K., 2012. Human-human reliance in the context of automation. Hum. Factors 54, 112–121. https://doi.org/10.1177/0018720811427034.

Macmillan, N.A., Kaplan, H.L., 1985. Detection theory analysis of group data: estimating sensitivity from average hit and false-alarm rates. Psychol. Bull. 98, 185. https://psycnet.apa.org/doi/10.1037/0033-2909.98.1.185.

Madhavan, P., Wiegmann, D., 2007a. Effects of information source, pedigree, and reliability on operator interaction with decision support systems. Hum. Factors 49, 773–785. https://doi.org/10.1518/001872007X230154.

Madhavan, P., Wiegmann, D., 2007b. Similarities and differences between human-human and human-automation trust: an integrative review. Theor. Issues Ergon. Sci. 8, 277–301. https://doi.org/10.1080/14639220500337708.

Mercado, J.E., Rupp, M.A., Chen, J.Y., Barnes, M.J., Barber, D., Procci, K., 2016. Intelligent agent transparency in human–agent teaming for Multi-UxV management. Hum. Factors 58, 401–415. https://doi.org/10.1177/0018720815621206.

Merritt, S.M., 2011. Affective processes in human–automation interactions. Hum. Factors 53, 356–370. https://doi.org/10.1177/0018720811411912.

Metzger, U., Parasuraman, R., 2005. Automation in future air traffic management: effects of decision aid reliability on controller performance and mental workload. Hum. Factors 47, 35–49.

Moray, N., 1986. Monitoring behaviour and supervisory control. In: Boff, K.R., Kaufman, L., Thomas, J.P. (Eds.), Handbook of Perception and Human Performance, Cognitive Processes and Performance, vol. 2. John Wiley & Sons, pp. 1–51.

Moray, N., Inagaki, T., 2000. Attention and complacency. Theor. Issues Ergon. Sci. 1, 354–365. https://doi.org/10.1080/14639220052399159.

Navon, D., Gopher, D., 1979. On the economy of the human processing system. Psychol. Rev. 86, 214–255.

Norman, D.A., Bobrow, D.G., 1976. On the analysis of performance operating characteristics. Psychol. Rev. 83, 508–510. https://psycnet.apa.org/doi/10.1037/0033-295X.83.6.508.

Panganiban, A.R., Matthews, G., Long, M.D., 2020. Transparency in autonomous teammates: intention to support as teaming information. J. Cogn. Eng. Decis. Mak. 14, 174–190. https://doi.org/10.1177/1555343419881563.

Parasuraman, R., Molloy, R., Singh, I.L., 1993. Performance consequences of automation-induced 'complacency. Int. J. Aviat. Psychol. 3, 1–23. https://psycnet.apa.org/doi/10.1207/s15327108ijap0301_1.

Parasuraman, R., Riley, V., 1997. Humans and automation: use, misuse, disuse, abuse. Hum. Factors 39, 230–253. https://psycnet.apa.org/doi/10.1518/001872097778543886.

Pearson, C.J., Geden, M., Mayhorn, C.B., 2019. Who's the real expert here? Pedigree's unique bias on trust between human and automated advisers. Appl. Ergon. 81, 102907 https://doi.org/10.1016/j.apergo.2019.102907.

Rieth, M., Hagemann, V., 2022. Automation as an equal team player for humans?-A view into the field and implications for research and practice. Appl. Ergon. 98, 103552 https://doi.org/10.1016/j.apergo.2021.103552.

Rosenthal, R., Rosnow, R.L., 1985. Contrast Analysis: Focused Comparisons in the Analysis of Variance. Cambridge University Press, Cambridge, England. https://doi.org/10.2307/2289419.

Roth, G., Schulte, A., Schmitt, F., Brand, Y., 2020. Transparency for a workload-adaptive cognitive agent in a manned–unmanned teaming application. IEEE Transact. Human-Mach. Syst. 50, 225–233. https://doi.org/10.1109/THMS.2019.2914667.

Rovira, E., Zinni, M., Parasuraman, R., 2002. Effects of information and decision automation on multi-task performance, 46. In: In*Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, pp. 327–331. https://doi.org/10.1177/154193120204600324.

Scerbo, M.W., 1996. Theoretical perspectives on adaptive automation. In: Parasuraman, R., Mouloua, M. (Eds.), Automation and Human Performance. Lawrence Erlbaum, Hillsdale, NJ, pp. 37–63.

Senders, J.W., 1983. Visual Scanning Processes. University of Tilburg Press, Netherlands.

Skraaning, G., Jamieson, G.A., 2021. Human performance benefits of the automation transparency design principle: validation and variation. Hum. Factors 63, 379–401. https://doi.org/10.1177/0018720819887252.

Stein, E.S., 1985. Atlantic City International Airport. In: Air Traffic Controller Workload: an Examination of Workload Probe (No. DOT/FAA/CT-TN82/24). Federal Aviation Administration Technical-Center, NJ.

Stowers, K., Kasdaglis, N., Newton, O., Lakhmani, S., Wohleber, R., Chen, J., 2016. Intelligent agent transparency: the design and evaluation of an interface to facilitate human and intelligent agent collaboration, 60. In: In*Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, pp. 1706–1710. https://doi.org/10.1177/1541931213601392.

Stowers, K., Kasdaglis, N., Rupp, M.A., Newton, O.B., Chen, J.Y., Barnes, M.J., 2020. The IMPACT of agent transparency on human performance. IEEE Transact. Human-Mach. Syst. 50, 245–253. https://doi.org/10.1109/THMS.2020.2978041.

Trapsilawati, F., Wickens, C.D., Qu, X., Chen, C.H., 2016. Benefits of imperfect conflict resolution advisory aids for future air traffic control. Hum. Factors 58, 1007–1019. https://doi.org/10.1177/0018720816655941.

Van de Merwe, K., Mallam, S., Nazir, S., 2022. Agent transparency, situation awareness, mental workload, and operator performance: a systematic literature review. Hum. Factors, 00187208221077804. https://doi.org/10.1177/00187208221077804.

Vagia, M., Transeth, A.A., Fjerdingen, S.A., 2016. A literature review on the levels of automation during the years. What are the different taxonomies that have been proposed? Appl. Ergon. 53, 190–202. https://doi.org/10.1016/j.apergo.2015.09.013.

Wickens, C.D., Dixon, S.R., 2007. The benefits of imperfect diagnostic automation: a synthesis of the literature. Theor. Issues Ergon. Sci. 8, 201–212. https://doi.org/10.1080/14639220500370105.

Wickens, C.D., Gutzwiller, R.S., Santamaria, A., 2015. Discrete task switching in overload: a meta-analyses and a model. Int. J. Hum. Comput. Stud. 79, 79–84. https://doi.org/10.1016/j.ijhcs.2015.01.002.

Wright, J.L., Chen, J.Y., Barnes, M.J., Hancock, P.A., 2017. The effect of agent reasoning transparency on complacent behavior: an analysis of eye movements and response performance, 61. In: In*Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, pp. 1594–1598. https://doi.org/10.1177/1541931213601762.